Biotechnology Principles & Processes

1. Which of the following is not a characteristic feature of pBR322 ?

(1) Size is about 4.3 kb

(2) It has two antibiotic resistance gene

(3) Low copy number

(4) It is chromosomal DNA

2. Given below are two statements:

Statement-I : Restriction enzyme belong to a larger class of enzymes called as nucleases.

Statement-II : Each restriction endonuclease recognise a specific palindromic nucleotide sequence in the DNA.

(1) Both Statement-I and Statement-II are incorrect.

(2) Statement-I is correct but statement-II is incorrect.

(3) Statement-I is incorrect but Statement-II is correct.

(4) Both Statement-I and Statement-II are correct.

3. Which of the following is true statement regarding DNA polymerase used in PCR ?

(1) It is isolated from plant cells

(2) It remains active at high temperature

(3) It is used to ligate foreign DNA in recipient cells

(4) It serves as a selectable marker

4. Restriction endonucleases are used in genetic engineering because: -

(1) They can degrade harmful proteins

(2) They can join DNA fragments

(3) They can cut DNA at variable site

(4) They can cut DNA at specific base sequences

5. Identify the correct set of statements :

(a) Origin of replication is responsible for initiating replication.

(b) DNA ligase acts on cut DNA molecules and joins their ends.

(c) Plasmid is autonomously replicating circular extra chromosomal DNA.

(d) More than 900 restriction enzymes that have been isolated from over 230 strains of bacteria.

Choose correct answer from options given below

(1) a and c only

(2) b, c and d only

(3) a, c and d only

(4) a, b, c and d all

6. The cloning vector with selectable marker ampR and tetR is

(1) YAC

(2) BAC

(3) pBR322

(4) λ-phage

7. How many DNA molecules will be produced from one DNA molecule after 6 PCR cycles ?

(1) 10 (2) 15

(3) 32 (4) 64

8. In gel electrophoresis, DNA fragments are separated on the basis of

(1) Molecular conformation

(2) Molecular charge

(3) Molecular size

(4) Both (1) and (3)

9. Which among the following vectors is called shuttle vector ?

(1) pBR322

(2) YEp

(3) λ-phage

(4) pUC 8

10. Which of the following recognition site is for EcoRI ?

(1) 5′–G↓AATTC–3′

3′–CTTAA↑G–5′

(2) 5′–AAGCTT–3′

3′–TTCGAA–5′

(3) 5′–GAATTC–3′

3′–CTTAAG–5′

(4) 5′–CTGCAG–3′

3′–GACGTC–5′

11. Select the mismatch

(1) Ori - responsible for initiating replication

(2) Sticky ends - single stranded free ends

(3) Genetic glue - T_4 ligase

(4) Alkaline phosphatase - removal of phosphate from 3′ end of DNA

12. Restriction endonucleases are found in

(1) Prokaryotes

(2) Yeasts

(3) All living organisms

(4) Bacteriophages

13. Which of the following is a palindromic sequence ?

(1) 5′-TGATCA-3′

3′-ACTAGT-5′

(2) 5′-CTGAGC-3′

3′-GACTCG-5′

(3) 5′-ATCGCC-3′

3′-TAGCGG-5′

(4) 5′-CGTAGA-3′

3′-GCATCT-5′

14. If EcoRI was used to cut the DNA during isolation of gene of interest, then vector DNA should be cut by using

(1) Bam HI

(2) Eco RI

(3) By any restriction endonuclease

(4) Pst I

15. If the gene of interest is inserted in ampicillin resistance gene in pBR322 then the recombinants will be selected by growing in medium containing

(1) Ampicillin

(2) Tetracycline

(3) Penicillin

(4) Both (1) and (2)

16. 3 molecules of DNA are amplified by PCR, then after 5 PCR cycles what will be total number of DNA molecules ?

(1) 96

(2) 64

(3) 32

(4) 35

17. Choose the option indicating correct sequence of following steps of RDT.

a. Isolation of gene of interest

b. Construction of rDNA

c. Selection of recombinants

d. Transformation of r-DNA into hosts

e. Culturing of recombinants in fermentor

(1) a → b → e → d → c

(2) a → b → d → c → e

(3) b → c → d → a → e

(4) e → c → b → d → a

18. A plasmid vector having 4 sites of EcoRI was subjected to restriction digestion by EcoRI. How many DNA fragments will be produced ?

(1) 2 (2) 3

(3) 4 (4) 5

19. Insertion of gene of interest in selectable marker leads to _____ of selectable marker select the option that fills the blank correctly

(1) Insertional activation

(2) Insertional inactivation

(3) Duplication

(4) Hyperactivation

20. Bacterial cells are made competent to take up alien DNA by

(1) Treating with specific concentration of divalent cation

(2) Heat shock

(3) Incubating cells in hot calcium chloride solution

(4) Incubating cells with armed retrovirus

21. Plasmids are used in genetic engineering because they are

(1) Easily available

(2) Able to replicate along with chromosomal DNA only

(3) Extra chromosomal DNA which can replicate without chromosomal DNA replication

(4) Rich in histone proteins

22. Alkaline phosphatase is used in recombinant technology which facilitates

(1) Removal of phosphate from 5′ end to prevent religation of vector

(2) Phosphorylation of terminal ribose

(3) Catalytic activity of DNA polymerase II

(4) Catalytic activity of taq polymerase

23. All the following is correct for vector except

(1) Presence of restriction sites for various restriction endonucleases

(2) Replicates independent of chromosomal DNA

(3) Presence of ori-site

(4) Replicates only with chromosomal DNA

24. Given below are two statements:

Assertion (A) : The first restriction endonuclease is Hind-II and cut specific sequence of six base pairs.

Reason (R) : Exonuclease remove nucleotides from the ends of DNA.

In the light of the above statements, choose the most appropriate answer from the options given below :

(1) Both (A) & (R) are correct but (R) is not the correct explanation of (A).

(2) (A) is correct but (R) is not correct.

(3) (A) is not correct but (R) is correct.

(4) Both (A) & (R) are correct and (R) is the correct explanation of (A).

25. Stained DNA is exposed to

(1) visible light

(2) UV light

(3) IR light

(4) Radio wave

26. Choose option with correct statement

A. r-DNA can be directly injected into the nucleus of an animal cell through microinjection

B. Upstream processing is one of the step of rDNA technology

C. Armed pathogen vectors are also used in transfer of rDNA into host

D. Transgenic models are used for studying new treatments for human diseases

(1) A and B only

(2) A, B and D

(3) B and D only

(4) C only

27. Given below are two statements:

Assertion (A) : Presence of insert results into insertional inactivation of the β-galactosidase gene.

Reason (R) : The colonies produce blue color, these are identified as recombinant colonies.

Choose the most appropriate answer

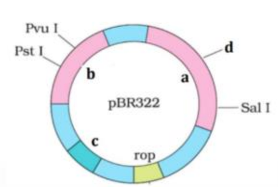
(1) Both Assertion and Reason are correct but Reason is not a correct explanation of the Assertion.

(2) Assertion is correct and Reason is not correct

(3) Both Assertion is not correct but Reason is correct

(4) Both Assertion and Reason are correct and the Reason is a correct explanation of the Assertion.

28. The figure below is diagrammatic representation of pBR322 which of the given options correctly identifies its certain components.



(1) a-tetR, b-ampR, c-ori, d-BamHI

(2) a-ampR, b-tetR, c-ori, d-BamHI

(3) a-tetR, b-ampR, c-BamHI, d-ori

(4) a-ampR, b-tetR, c-PvuII, d-EcoRI

29. Match column-I with column-II.

Column-I	Column-II
(a) Microinjection	**(i) Competent cells**
(b) Biolistics	**(ii) Plant cells**
(c) Heat shock	**(iii) Bacterial cells**
(d) Treatment with Ca2+	**(iv) Animal cells**

Choose the most appropriate answer from the options given below :

(1) a-(iv), b-(ii), c-(iii), d-(i)

(2) a-(ii), b-(iv), c-(i), d-(iii)

(3) a-(iv), b-(i), c-(ii), d-(iii)

(4) a-(i), b-(ii), c-(iii), d-(iv)

30. The key tools for recombinant DNA technology are

(1) Restrication enzyme, polymerase, hydrolase, vectors

(2) Recognition enzyme, polymerase, ligase, vector

(3) Restriction endonuclease, polymerase, ligase, vector

(4) Restriction enzyme, polymerase, dehydrogenase vector

31. Which amongst the following are protein based modern methods that ensure detection of HIV ?

(a) PCR

(b) ELISA

(c) Western blotting

(d) Northern blotting

Choose the correct answer from options given below

(1) (a) and (b)

(2) (b) and (c)

(3) (a) and (d)

(4) (b), (c) and (d)

32. Identify the correct set of statements :

(a) When different DNAs are cut by same restriction enzyme, the resultant DNA fragments have the same kind of sticky ends.

(b) The seperated bands of DNA are cut out from the agarose gel and extracted from the gel piece. This step is known as elution.

(c) The RNA can be removed by treatment with protease.

(d) If any protein encoding gene is expressed in a heterologous host it is called a recombinant protein.

Choose the correct answer from options given below :

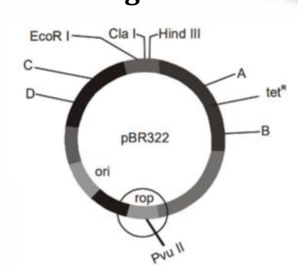
(1) a and b only

(2) b, c and d only

(3) a, b and d only

(4) a and d only

33. Structure of plasmid vector is shown in the diagram with its specific antibiotic resistance genes and recognition sites of particular RE.



Select the correct match

(1) A - BamH III, used to remove tetracycline resistant gene

(2) D - When used as site for inserting gene of interest, then recombinant organism can grow in tetracycline medium.

(3) C - When used as site for inserting alien DNA, then recombinant form will grow in both ampicillin and tetracycline medium.

(4) B - Sal-I which removes the gene coding for producing β-galactosidase.

34. Non-recombinants transformants will

(1) Grow in ampicillin and tetracycline both

(2) Grow in ampicillin but not tetracycline

(3) Grow in tetracycline but not ampicillin

(4) Grow neither in tetracycline nor in ampicillin

35. Bacterial host cells are made competent to take up rDNA by-

(1) Treating with Na+

(2) Treating with Al 3+

(3) Treating with Ca2+

(4) More than one options

36. The construction of first r-DNA emerged from the possibility of linking a gene encoding antibiotic resistance with native plasmid of :-

(1) Salmonella typhimurium

(2) E. coli

(3) Yeast

(4) Thermus aquaticus

37. Given below are two statements:

Statement-I : Restriction enzyme belong to a larger class of enzymes called as nucleases.

Statement-II : Each restriction endonuclease recognise a specific palindromic nucleotide sequence in the DNA.

Choose the most appropriate answer

(1) Both statement I and II are correct.

(2) Statement I is correct but statement II is incorrect.

(3) Statement I is incorrect but statement II is correct.

(4) Both statement I and II are incorrect.

38. Restriction enzyme cuts DNA

(1) between same two bases on opposite strands, randomly DNA.

(2) between same two bases on opposite strands, a little away from centre of DNA sequence recognized.

(3) between two different bases on opposite strands, in centre of DNA sequence recognized.

(4) between two different bases on opposite strands, little away from centre of DNA sequence recognized.

39. Select true(T) and false (F) w.r.t PCR

A. Taq DNA polymerase remains active at high temperatures

B. A pair of primers with exposed 5'-OH groups will bind to the complementary sequence of template DNA

C. Annealing of primer with the template DNA requires 95°C

D. PCR can be used to detect a specific disease causing mutation or pathogens present in very low concentrations

(1) A-T, B-F, C-F, D-T (2) A-T, B-F, C-T, D-F

(3) A-F, B-T, C-F, D-T (4) A-F, B-T, C-T, D-F

40. For alien DNA to replicate it needs to be a part of:

(1) chromosome without origin of replication site

(2) mitochondrial DNA with origin of replication site

(3) Vector with origin of replication site

(4) cytoplasmic DNA with origin of replication site

41. Which of the following is correct?

(1) Any piece of DNA linked to ori site will be replicated inside host.

(2) Number of replication copies is under control of recognition site.

(3) Vector should not be chosen based on number of copies supported by it

(4) More than one option.

42. Arrange the following steps in correct sequence resulting in transformation process

(A) Bacterial cells with recombinant DNA are incubated on ice

(B) Placing them briefly at 42°C

(C) Putting bacterial cells with recombinant DNA back-on ice.

(D) Bacterial cells are treated with a specific concentration of a divalent cation, such as calcium.

(1) A → B → C → D

(2) A → B → D → C

(3) D → A → B → C

(4) D → A → C → B

43. In order to induce the uptake of plasmids by bacteria, the bacteria are made competent by treating with

(1) Ice cold calcium chloride

(2) Heat shock at 42°C and then putting them back on ice

(3) Sodium chloride at 4°C

(4) Bromophenol blue

44. The colonies of recombinant bacteria appear white in contrast to blue colonies of non-recombinant bacteria due to

(1) Insertional inactivation of α-galactosidase in non-recombinant bacteria

(2) Activation of γ-galactosidase in recombinant bacteria

(3) Inactivation of glycosidase enzyme in recombinant bacteria

(4) Inactivation of β-galactosidase gene in recombinant bacteria

45. How many cycles of PCR are required for obtaining 512 molecules of DNA if your starting material was one molecule of DNA ?

(1) 8

(2) 9

(3) 256

(4) 511

46. Read the statements given below and mark the incorrect one with respect to DNA electrophoresis

(1) Smaller the fragment size, farther it travels from anode

(2) DNA fragments run from cathode to anode

(3) Larger DNA fragments run slowly than smaller fragments

(4) Methylene blue dye is used to colour DNA before gel loading smaller fragments

47. A bacterium adds methyl groups to its DNA, by a process known as modification, in order to

(1) Clone its DNA

(2) Turn its gene 'on'

(3) Transcribe many genes simultaneously

(4) Protect its DNA from its own restriction enzymes

48. A scientist tried to clone a gene of interest into Pst I site of plasmid pBR322...

(1) Recombinants

(2) Only non recombinants

(3) Non transformants

(4) Both (1) and (2)

49. Given below are two statements :

Statement I : Restriction endonuclease is most important tools of biotechnology.

Statement II: Ligase enzyme is responsible for joining of DNA fragement.

In the light of the above statements, choose the correct answer from the options given below :

(1) Both statement-I and statement-II are incorrect

(2) Statement-I is correct but statement-II is incorrect

(3) Statement-I is incorrect but statement-II is correct

(4) Both statement-I and statement-II are correct

50. Which technique is useful for marking many copies of DNA :

(1) Gene gun

(2) PCR

(3) ELISA

(4) Micro propagation

Answer Key

1. (4)
2. (4)
3. (2)
4. (4)
5. (4)
6. (3)
7. (4)
8. (4)
9. (2)
10. (3)
11. (4)
12. (1)
13. (1)
14. (2)
15. (4)
16. (1)
17. (2)
18. (3)
19. (2)
20. (1)
21. (3)
22. (1)
23. (4)
24. (1)
25. (2)
26. (2)
27. (2)
28. (1)
29. (1)
30. (3)
31. (2)
32. (3)
33. (2)
34. (1)
35. (3)

36. (1)
37. (4)
38. (2)
39. (1)
40. (3)
41. (1)
42. (3)
43. (1)
44. (4)
45. (2)
46. (1)
47. (4)
48. (2)
49. (4)
50. (2)